Guidelines for the Description of Barley Germplasm Resources

1 Scope

This specification establishes the descriptors for barley germplasm resources and their grading criteria.

These guidelines apply to the collection, organization, and preservation of barley germplasm resources; the development of data standards and data quality control specifications; and the establishment of databases and information-sharing network systems.

2 Normative References

The provisions in the following documents shall form part of this specification by reference therein. For dated reference documents, all subsequent corrigenda (excluding errata) or revised editions shall not apply to this specification. However, parties to agreements concluded in accordance with this specification are encouraged to consider whether the latest editions of these documents may be used. For undated reference documents, the latest edition shall apply to this specification.

ISO 3166 Codes for the Representation of Names of Countries

GB/T 2659: Country and Region Name Codes

GB/T 2260: Administrative Division Codes of the People's Republic of China

GB/T Code for Unit Affiliation

GB4404.1—1996: Seeds of cereal crops

GB/T 3543.4—1995: Test for germination in the inspection of crop seeds

GB5519: Method for Determining the Thousand-Kernel Weight of Grains and Oilseeds

GB7416 Beer Barley

GB10367 Feed Barley

GB11760 Barley

GB/T 15796—1995 Specifications for the Monitoring and Investigation of Fusarium Head Blight in Wheat Crops

GB2905—82 Method for Determination of Crude Protein in Seeds of Cereal and Leguminous Crops (Semi-Kjeldahl Nitrogen Determination Method); GB/T5511—1985 Method for Determination of Crude Protein in Grains and Oil Crops

GB/T 5514—1985 — Testing of Grains and Oilseeds: Method for Determination of Starch

GB4801—84 Determination of Lysine in Grain Crops: Dye-Bound Lysine (DBL) Method

GB7649—87 Pre-treatment for the Determination of Amino Acids in Grain Crops

3 Terms and Definitions

3.1 barley

Gramineae; genus *Hordeum*; common barley, an annual herbaceous plant, scientific name *Hordeum vulgare* L., with a chromosome number of 2n=2X=14.

3.2 Barley Germplasm Resources

Wild resources of barley, local varieties, selected and bred varieties, lines, genetic materials, and so on.

3.3 essential information

Descriptive information on the basic status of barley germplasm resources, including the unified identifier, germplasm name, scientific name, and place of origin,

Germplasm types, etc.

3.4 Morphological Characteristics and Biological Properties

The phenological stage, botanical morphology, yield traits, and other characteristic features of barley germplasm resources.

3.5 quality characteristic

The malt processing quality, beer brewing quality, and nutritional quality of barley germplasm resources, among other characteristics; these include grain texture, plumpness, husk retention rate, water sensitivity, starch content, protein content, *β*-glucan content, extract content, and saccharification capacity.

3.6 Stress Resistance

The adaptability or resistance of barley germplasm resources to various abiotic stresses, including soil and climatic factors, encompasses cold tolerance, drought resistance, moisture tolerance, salt tolerance, and acid tolerance.

3.7 Resistance to pests and diseases

The adaptability or resistance of barley germplasm resources to various pest and pathogen stresses, including yellow leaf disease, yellow dwarf disease, Fusarium head blight, powdery mildew, stripe disease, root rot, and aphids.

4 Base Situation

4.1 Nationally Unified Numbering

The unique identifier for each barley germplasm accession consists of three uppercase Chinese Pinyin letters followed by a five-digit sequential number. 4.2 Germplasm Bank Numbering

The accession number assigned to each barley germplasm entry in the National Long-term Crop Germplasm Resource Bank consists of "I1E" followed by a five-digit sequential number.

4.3 Introduction Number

The designation assigned to barley germplasm when it was introduced from abroad.

4.4 Collection ID

The original identification number assigned during the field collection of barley germplasm.

4.5 Germplasm Name

The Chinese name for barley germplasm.

4.7 Genetic resource foreign name

The original foreign name of the imported barley germplasm from abroad; the Chinese pinyin name of the domestic barley germplasm.

4.7 Field Name

Gramineae.

4.8 generic name

The genus Hordeum (*Hordeum).*

4.9 scientific name

Common barley (*Hordeum vulgare* L).

4.10 country of origin

The name of the country, region, or international organization to which the origin or collection site of the barley germplasm belongs.

4.11 Country of Origin

The name of the administrative province, municipality directly under the central government, or autonomous region to which the place of origin or collection of domestic barley germplasm belongs; the name of the national first-level administrative region of origin for germplasm introduced from abroad.

4.12 **country of origin**

The names of the counties, townships, and villages of origin for domestic barley germplasm resources.

4.13 height

The elevation of the origin or collection site of the barley germplasm. The unit is m.

4.14 longitude

The longitude of the origin or collection site of the barley germplasm. The unit is () and (). The format is DDDFF, where

Use DDD as the primary framework and FF as a supplement.

4.15 latitude

The latitude of the origin or collection site of the barley germplasm. The unit is () and (). The format is DDFF, where

DD represents the degree, and FF represents the fraction.

4.16 **Origin**

Names of the provinces and counties where domestic barley germplasm originated; names of the countries, regions, or international organizations from which imported barley germplasm was introduced.

4.17 **Preserving Entity**

The original name of the institution where the barley germplasm was preserved prior to its submission to the National Long-term Crop Germplasm Resource Bank.

4.18 **Save Unit Number**

The original custodian assigned a accession number to the barley germplasm.

4.19 **Genealogy**

Parent lines and hybridization combinations for selected barley varieties (lines).

4.20 Breeding Unit

Name of the institution or individual responsible for the breeding and development of barley varieties (lines).

4.21 Years of age

Year in which the barley variety (line) was successfully developed.

4.22 Breeding Method

The breeding methods employed for developing barley varieties (lines).

4.23 **Genetic Material Type**

Barley germplasm types are classified into the following six categories.

1 Wild resources
2 local variety
3 Selected breeding varieties
4 Strain
5 genetic stocks
6 other

4.24 Image

Image filename for barley germplasm. Format:.jpg.

4.25 observation place

Name of the observation site for the morphological characteristics and biological traits of barley germplasm.

5 Morphological Characteristics and Biological Properties

5.1 Sowing Time

For the field sowing date used in the characterization and trait identification of barley germplasm resources, express it as "year month day" in the following format:

“YYYYMMDD”。

5.2 Period of Emergence

The date on which more than 50% of the seeds have their coleoptiles emerging above the soil surface by 1 cm shall be expressed as "year month day" in the format "YYYYMMDD".

5.3 Heading Period

The date on which more than 50% of the stem spikes have emerged through the leaf sheath 1 cm above the base (excluding the awn) shall be indicated as "year month day," in the following format:

“YYYYMMDD”。

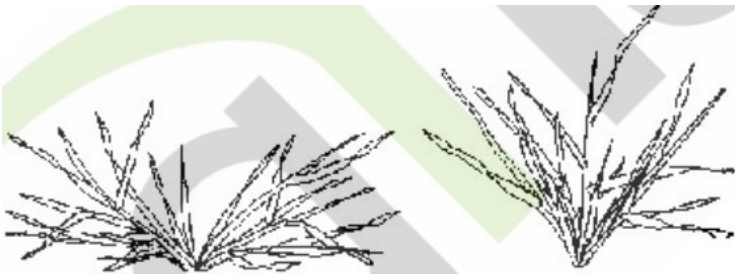
5.4 Maturation Period

More than 80% of the plants exhibit normal yellowing and withering, and the grains become hard; the date on which they can no longer be cut with a fingernail shall be expressed as "year month day" in the format "YYYYMMDD".

5.5 Seedling Growth Characteristics

The growth status of barley germplasm seedlings is classified into the following three types.

1 crawl
2 semi-prostrate
3 up-right



crawl up-right

Figure 1: Growth Habits of Seedlings

5.6 Winter–spring nature

The low-temperature vernalization requirements of barley germplasm.

1 winterness
2 semiholism
3 spring habit

5.7 photoperiod response

The light-length response characteristics of barley germplasm.

1 slow
2 secondary
3 sensitive

5.8 Branching capacity

The number of tillers per individual barley germplasm plant.

1 Strong
2 China
3 weak

5.9 Color of the auricles

The color of the auricles between the leaves and leaf sheaths of barley germplasm.

1 white
2 Green
3 red
4 purple

5.10 Blade length

Abbreviated as leaf length; it refers to the length from the tip of the first leaf to the leaf sheath. The unit is cm.

5.11 Blade Width

Abbreviated as leaf width, it refers to the width of the middle part of the first leaf; the unit is cm.

5.12 Blade Posture

Abbreviated as Ye Zi; the spatial growth habit of the leaves after the barley seed material has reached heading stage.

1 up-right
2 Flat
3 hang down

5.13 Leaf color

Abbreviated as leaf color, it refers to the color of the leaves during the heading stage of barley germplasm.

Work: Light Green

2 Green
3 dark green

5.14 waxy stem and leaves

The amount of white waxy deposits on the stems and leaves of barley germplasm.

Work is absent or minimal
2 secondary
3 More

5.15 Plant Type

Plant growth habit of barley germplasm after jointing and heading.

1 compact
2 semi-compact
3 loose

5.16 Local ripening type

The degree to which barley germplasm matures earlier or later than local mid-season varieties.

Industrial exceptionally early-maturing

2 precocity
3 Medium-mature
4 late-maturing
5 Extremely late maturing

5.17 Plant Height

The height from the tiller node of the main stem to the tip of the panicle in mature plants, excluding the awn; expressed in cm.

5.18 Stem Diameter

Diameter at the midpoint of the second internode above ground of the main stem. Unit: mm.

5.19 Number of spikes per plant

Average number of spikes per plant in barley germplasm; the unit is spikes.

5.20 Sui Zi

The attachment posture of the spikelets on the stem when they mature (see Figure 2).

1 up-right
2 horizontal
3 hang down



Figure 2:穗姿

5.21 Suihe Mangse

The color of the panicles and awns when the barley germplasm has reached maturity.

1 Huang
2 Gray
3 purple
4 brown
5 black

5.22 Prismatic

Based on the grain-setting performance of the three-rowed spikelets on the rachis, the ridge types of barley germplasm are classified into three types (see Figure 3).

1 Bilateral
2 intermedius type
3 Hexagonal



Figure 3: Prism Type

5.23 穗 Length

Length from the base of the spike to the tip of the spike (excluding the awn). Unit: cm.

5.24 spike density

The density of triple spikelet arrangement on the spike axis of barley germplasm. Based on the number of spikelet nodes within the spike axis over a 4 cm segment in the middle of the spike, the material is classified into three categories (see Figure 4):

1 rare
2 secret
3 Extremely dense



Figure 4: Spike Density

5.25 Sui Branch

Whether secondary spikelets develop on the spike axis or on the triple spikelets of barley germplasm.

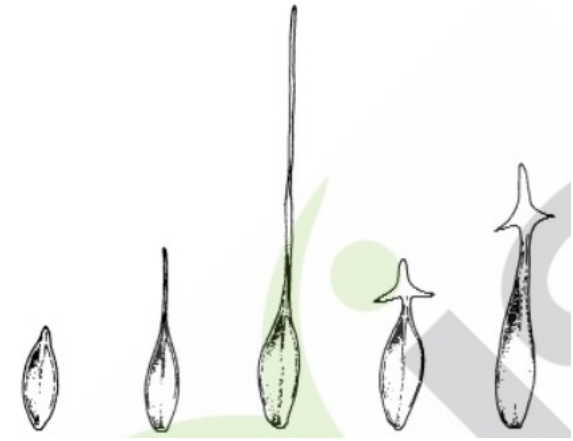
1 Non-branching
2 branch

5.26 Mang Type

The length and shape of the awns on the central and lateral spikelets of the panicle. The described types are divided into 16 categories (see Figure 5).

0 Barbed-free
1 Micro芒
2 Equisetum glaucescens
3 Short awn
4 Long awn
5 The middle and long sides are glabrous
6 Medium to long lateral microspikes
7 Medium to long lateral bracts, short awns
8 Short to medium lateral glumes without awns
9 No spines on the mid-sides
10 No neck hook margin
11 short hook margin
12 Long hooked edge
13 Medium-length hook, side short hook point

14	The middle hook margin is slightly acute
15	The hooked side is glabrous



Barbed-free short barbed, long barbed short hooked barbed, long hooked barbed

Figure 5:芒-type

5.27 Mang nature

Whether the spikelets of barley germplasm possess dentate protrusions is shown in Figure 6.

1 dental bristles
2 rays of light

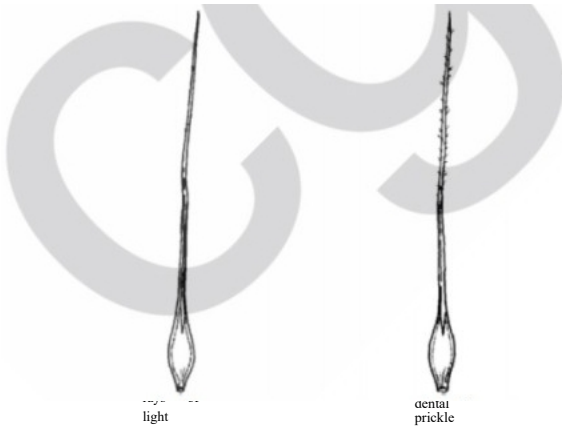


Figure 6: Manic Nature

5.28 Lateral Spikelet

The degree of developmental degeneration of lateral spikelets in the germplasm of two-awned barley.

1 Lateral
2 No lateral

5.29 Protect the width and narrowness of the girdle

The width of the middle part of the glume in barley germplasm.

1 narrow
2 **wide**

5.30 穗轴茸毛

The length of the pubescence on the rachis of the threshed spikelets (see Figure 7).

1 short
2 long



Short long

Figure 7: Glume Hairs on the Rachis

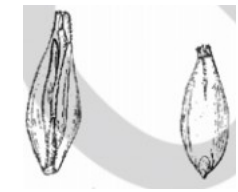
5.31 Number of Grain per Ear

Abbreviated as "grain count per spike," it refers to the actual number of grains produced by each spike. The unit is grains.

5.32 Encapsulated Nature

After seed maturation, does it naturally detach from the glume (see Figure 8)?

1 Pi
2 Naked



Coarse barley hull-less barley

Figure 8 Shell-like property

5.33 Grain Color

Color of mature grains in barley germplasm.

1 Huang
2 Lan
3 **purple**
4 **brown**
5 **black**

5.34 Grain Shape

Briefly referred to as grain morphology, the shape of mature grains in barley germplasm is classified into four types (see Figure 9).

1 Oblate
2 orbicular-ovate
3 oval
4 circular



Figure 9: Grain shape

5.35 **1,000-grain weight**: the weight of 1,000 naturally air-dried barley grains. Unit: g.

5.36 Seed dormancy period

The time required for barley germplasm to develop from maturity to the point at which the seeds can germinate normally. The unit is d.

5.37 Germination Percentage

The proportion of barley seeds that germinate under suitable temperature and humidity conditions, expressed as a percentage (%).

6 Quality Characteristic

6.1 Grain fullness

Briefly referred to as fullness, it denotes the screening rate of barley grains through a 2.5 mm long-hole sieve, expressed as a percentage.**6.** 2. Seed coat retention rate

The hull percentage, abbreviated as hull ratio, refers to the proportion of the hull weight in the total weight of hull barley grains. It is expressed as a percentage (%).

6.3 Grain texture

Abbreviated as grain substance, it is classified according to the proportion of vitreous material in the endosperm of the grain as follows:

1 stereoplasm
2 semi-rigid
3 Soft

6.4 content of starch

The proportion of crude starch in the weight of barley grains, expressed as a percentage.

6.5 Protein content

The proportion of protein in the weight of barley grains, expressed as a percentage (%).

6.6 Lysine Content

The proportion of lysine in barley grains per unit weight, expressed as a percentage.

6.7 Extracts

A specified fineness and unit weight of barley malt powder are saccharified under defined temperature conditions to produce a quantity of water-soluble substances, expressed as a percentage.

6.8 β -glucan content

The proportion of β -glucan in the weight of barley grains, expressed as a percentage.

6.9 saccharifying power

At a temperature of 20 °C, the ability of 100 g of anhydrous barley malt to hydrolyze soluble starch into maltose within 30 min. The unit is W·K (1 W·K is defined as the production of 1 g of maltose).

6.10 Hydrophilicity

A physiological response that inhibits germination in barley seeds due to excessive moisture during the germination process.

1 not have
3 Extremely lightweight
S Mild
7 moderate
9 serious

7 Resilience

7.1 winter resistance

The ability of barley germplasm to tolerate or resist low-temperature damage.

1 highly resistance
3 resist
5 moderate resistance
7 Not resistant
9 Extremely low resistance

7.2 Drought Resistance

The ability of barley germplasm to tolerate or resist soil drought.

Industrial High-Durability
3 be able to bear or endure
5 Zhongnai
7 intolerance
9 Extremely intolerant

7.3 moisture-proof

The ability of barley germplasm to tolerate or resist waterlogging damage.

1 High durability
3 be able to bear or endure
5 Zhongnai
7 intolerance
9 Extremely intolerant

7.4 salt tolerance

The ability of barley germplasm to tolerate or resist excessive salt content in the soil.

1 High durability
3 be able to bear or endure
5 Zhongnai
7 intolerance
9 Extremely intolerant

7.5 Acid resistance

The ability of barley germplasm to tolerate or resist soil acid stress.

1 High durability
3 be able to bear or endure

5 Zhongnai

7 Not resistant

9 Highly susc-

eptible to diseases and pests; 8.

Disease and pest resistance

8.1 Resistance to yellow leaf disease

Barley yellow mosaic disease is caused by the Barley yellow mosaic virus (*Barley yellow mosaic virus,* BaYMV) and the Barley mild mosaic virus (*Barley mild mosaic* virus, BaMMV); it is a major disease transmitted by the soil-borne grass polymyxin (*polymyxa graminis*) as a vector. The resistance levels of barley germplasm to this disease are ranked as follows:

1 highly resistance(HR)
3 resist(R)
5 moderate resistance(MR)
7 feel(S)
9 High Sensitivity (HS)

8.2 Resistance to yellow dwarf disease

Barley yellow dwarf disease is a major disease of barley caused by the yellow dwarf virus (*Barley yellow dwarf virus,* BYDV). The resistance of barley germplasm to this disease is graded as follows:

1 resist(R)
3 be able to bear or endure(T)
5 MT
7 feel(S)
9 High Sensitivity (HS)

8.3 Fusarium head rot resistance

Barley Fusarium head blight is a major disease of barley caused by Fusarium *graminearum*. The resistance levels of barley germplasm to this disease are classified as follows:

1 highly resistance(HR)
3 resist(R)
5 moderate resistance(MR)
7 feel(S)
9 High Sensitivity (HS)

8.4 Resistance to powdery mildew

Powdery disease is the primary disease of barley caused by the barley powdery mildew pathogen (*Erysiphe graminisf* .sp. *Hodei*). The resistance levels of barley germplasm to this disease are classified as follows:

1 High resistance (HR)
3 resist(R)
5 China Anti-MRSA
7 feel(S)

9 High Sensitivity (HS)

8.5 Resistance to stripe disease

Barley stripe disease is a major disease of barley caused by the barley stripe disease pathogen (Pyrenophora graminea). The disease resistance of barley germplasm can be described as follows:

1 highly resistance(HR)
3 resist(R)
5 moderate resistance(MR)
7 feel(S)
9 High Sensitivity (HS)

8.6 Resistance to root rot

Barley root rot is caused by the pathogen Helminthosporium sativum and Fusarium spp. The resistance levels of barley germplasm are classified as follows:

1 highly resistance(HR)
3 resist(R)
5 moderate resistance(MR)
7 feel(S)
9 High Sensitivity (HS)

8.7 Resistance to net blotch disease

Barley leaf spot is caused by the ascomycete Pyrenophora teres of the genus Pyrenophora and primarily affects the leaves.
The resistance levels of barley germplasm are classified as follows:

1 highly resistance(HR)
3 resist(R)
5 China Anti-MRSA
7 feel(S)
9 High Sensitivity (HS)

8.8 Aphid resistance

The aphids that primarily危害 barley in China include the binary aphid (Toxoptera graminium Rond.), the long-tubular aphid *(Macrosiphum granarium Kiry), and the rice aphid (Rhopalosiphum padi Linn.). The* resistance levels of barley germplasm to aphids are classified as follows:

1 highly resistance(HR)
3 resist(R)
5 moderate resistance(MR)
7 feel(S)
9 High Sensitivity (HS)

9 Other Characteristics and Features

9.1 Main Application

Primary uses of barley grains.

1 Grain
2 forage

3 beer

9.2 Mutant

Specialized trait mutations in barley arising spontaneously or through mutagenesis.

9.3 Nearly isogenic line

A specialized genetic material in barley that exhibits only a single genetic variation.

9.4 Morphological Marking

A marker trait gene named according to the International Barley Gene Nomenclature.

9.5 chromosome disorder

Variation in the chromosome number, morphology, and structural characteristics of barley germplasm.

9.6 Biochemical Marker

Protein and enzyme markers in barley germplasm.

9.7 molecular markers

DNA molecular markers for the major traits of barley germplasm include restriction fragment length polymorphism (RFLP), random amplified fragment polymorphism (RAPD), amplified fragment length polymorphism (AFLP), simple sequence repeat polymorphism (SSR), and single nucleotide polymorphism (SNP), among others.

9.8 remarks

A detailed description or code-specific explanation for the barley special descriptor.